

Graphic 2D-Inspired Characters in *The Bad Guys*

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Figure 1: A selection of shots requiring different techniques to meet our illustrative aesthetic

ABSTRACT

The world of *The Bad Guys* is an homage to illustration and our characters take inspiration from elements of hand-drawn 2D animation. In this talk we present various challenges of developing graphic characters with solutions that scale for feature production.

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1 FACIAL ACCENTUATIONS

In 2D animation, linework is often introduced on characters to heighten expressions and enhance the graphic look. In 3D animation, previous techniques constrain lines to the head and animators manually modify the shape. In *The Bad Guys*, our *Expression lines* exist and move directly on a character's skin. This yields more automated deformation and avoids misregistration on camera moves. Lines are rigged via normal scaling and animators pose in UV space with transparency control. Faces have up to 18 rigged lines and placement is integrated into characters' library poses, but are overridden as necessary for performance.

Our show has the added complexity that these lines exist on characters with hair/fur. If 1) we render the lines directly, the lines

would be occluded by fur or if 2) we lift lines off the surface, we encounter problematic areas on grazing angles and creases (eg. snarls/squints). Alternatively, rendering the lines indirectly (using closest point or follicle position) results in a distorted or spatially-shifted line. Instead, at render-time we extrude a volume from the line towards the camera and sample the volume mask onto the fur. The line is effectively embedded in the fur and is identical to the posed graphic shape that the animator intended.

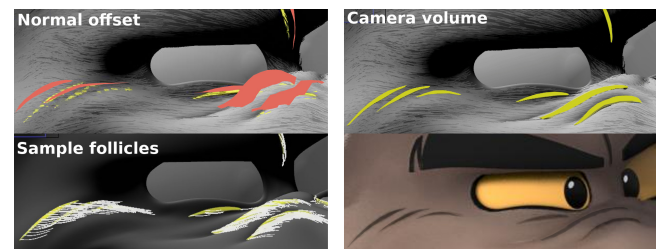


Figure 2: Left: issues with expression lines and fur. Right: camera-based volume mask maintains graphic line quality

A large advantage to a shader-based technique is that integration is significantly improved (same material properties as surrounding fur, introduce drawn-line texture, etc). Note for fast moving geometry/cameras, renderers typically output a blurred position (across the time-step), which should be transformed correctly prior to sampling the volume. To minimize computation, we employ volume caching considering upstream character animation and tool settings. This technique proved highly scalable (used on 5200+ characters in the film) with animators driving dynamic shaders with no downstream intervention. When modifications were necessary, we implemented tools for render-time editing (position,

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thickness, opacity, etc) as well as ID & coverage AOVs for grading in compositing.

Many additional facial techniques were developed to meet specific artistic challenges. *Blush* and *Disheveled* effects were executed similarly to *Expression lines*, though the render-time volumes were generated from painted UDIM textures. *Eyeliner* lines sit in the crevice of the eye, rigged to automatically adjust line-weight as a character's gaze shifts (thicker line further from the eye center). *Eye circles* are lines that span a partial circumference about the pupil, utilized when the character is enraged or surprised. Characters were also rigged with multiple sets of teeth (including *zigzags* and *plates*), swapped in per artistic demand.



Figure 3: 2D strokes converted into 3D geometry, automated process tracks corresponding USD asset at the proper depth

2 PERFORMANCE LINES

Stylistically, it was important to augment character performance with hand-drawn linework with its endearing imperfections. To facilitate this, we extended our 3D animation tool *Premo* with 2D animation capabilities. Animators draw strokes of varying pressure, thickness and transparency, which could be animated, transformed, and tracked. Elements typically fall into these categories:

- *Speed lines*: Smears of fast-moving geometry to marry neighboring poses, which would otherwise strobe in motion (a stylistic choice was to use a short camera shutter)
- *Multi-limbs*: Animating with multiples is a traditional 2D animation technique to convey speed
- *Impacts*: fast-evolving shapes to exaggerate/amplify contact
- *Misc. effects*: sweat, split, complex facial lines, electricity, etc

A drawing's name drives an automated process searching for a matching USD asset to place the strokes correctly in depth (via perspective transformation and updating corresponding stroke properties). The strokes are converted to a matching geometric and shading representation and installed in USD. 3D stroke representation proved useful for complex layering, animation tracking, and slight camera adjustments (without requiring a redo of the hand-drawn animation).

Interactively, artists could draw on 3D animated geometry and/or on pre-rendered plates from any department. The latter was useful with *multi-limbs* and *smears*, where exact color-matching was critical, though careful color-transform tracking throughout was necessary to ensure a match.

On previous shows, executing this type of work was tedious, split across departments, and required many meetings and iterations. This approach unified 2D & 3D in one application, providing large efficiency gains and creative improvements in 400+ shots.

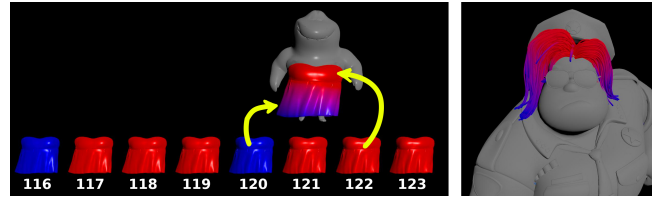


Figure 4: Stepped simulation blending in cloth & hair

3 STEPPED SIMULATION

Our characters are animated with poses changing every frame (ie. animated on 1's), yet we wanted to utilize the traditional technique of held frames. This change of cadence is an artistic choice that accentuates our graphic poses. While previous films simply held entire frames to give a stepped appearance, we wanted the control to step all or parts of a character at different held rates via weights that can vary spatially & temporally.

For example, this level of control is important for hair & cloth simulations that are stepped on 4's (Figure 4), but attached to a character animated on 1's. Simply holding the hair & cloth would detach it from the corresponding character on intermediate frames. Hence, the stepped frame needs to be properly deformed into the current frame's space.

Often, these step weights are determined via proximity or painted, but at times the step size is also driven via distance to camera, as holding the silhouette shape is typically more important than internal motion as a character recedes in space.

4 STYLIZED LIGHTING

Physical shading models provide rich internal detail, which conflicts with how a traditional artist illustrates (eg. clean silhouettes, smooth gradients, selective detail). To reduce the visual complexity of hair/fur, we utilize a diffuse hair shading model (Kajiya-Kay) and subsurface scattering. This reduces overall hair-to-hair shadow detail while still providing Lighting a robust occlusion method. We further coalesce high-frequency information into painterly strokes with a custom Kuwahara-based anisotropic kernel.

To flatten our characters' shading to appear more graphic, we blend in the diffuse albedo AOV graded to match the lighting conditions. This helps remove volumetric shading and occlusion, as well as restore local color to the characters. To add subtle detail in areas of low contrast, an in-render stylized line drawing AOV (derived from geometry silhouette/creases) was utilized, scoped opposite the key light direction.

Rim lighting is an important component for accentuating our graphic poses. First, a render-time operator would override all of our characters' materials with a white diffuse shader to ensure a consistent light response. Artists further sculpted the rim light AOV in compositing (reminiscent of a drawn line) with additional directional control and core roll-off.

Further, artists would modulate light AOVs with hatching or texture to add additional hints of selective detail. The effect was dialed in compositing with texture provided from a number of sources, including painted maps, sourcing a library of images for projection, or procedurally-generated lines.